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DOCUMENT SPLITTING TOOL

Abstract

Various embodiments disclosed relate to automated docketing of incoming electronic communications and documents. The present disclosure includes methods and systems for identifying omnibus documents containing more than one event, and splitting those omnibus documents into the individual events or documents.

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Background/Summary

CLAIM OF PRIORITY [0001] This application is a continuation of and claims the benefit of priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 17/723,121, filed on Apr. 18, 2022, which claims the benefit of priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application Ser. No. 63/232,819, filed on Aug. 13, 2021, each of which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] For a given project, a variety of deadlines and tasks may be required. Often, a large number of incoming communications and documents have to be sorted and analyzed for those tasks. When the project is collaborative, such as with an outside firm or company, communication between the outside firm and the internal project management can be required to properly docket deadlines, tasks, or other actions that should be met throughout the course of the project. Often, this aspect of project management is done through free form communication and human interpretation of instructions from the outside firm.

[0003] In a business, docketing systems can be used to track workflow deadlines, tasks, and progress. For example, in a given project, a docketing system can track when the project is opened, what initial filings are due when, and correspondence coming in regarding that project. The docketing system can additionally be used to record deadlines and flag correspondence or documents that may require a response. However, in a general docketing setting, a large number of documents are incoming and often need to be sorted during docketing. Those documents can come in a variety of file formats, and can sometimes be grouped together, requiring more work once received to efficiently split and docket each document.

SUMMARY OF THE DISCLOSURE

[0004] Disclosed is a method of sorting and separating incoming electronic communications and documents for project management and docketing. The method takes incoming documents, such as in portable document format (PDF) form, and can assist in separating of multi-document packages into individual events or items.

[0005] Incoming documents can sometime be received in packets or packages containing more than one individual document. For example, an incoming electronic communication can include an attached PDF file with multiple documents, each document indicating a different event in the project or file history.

[0006] In this case, where multiple distinct documents are contained in one electronic file, it is helpful to separate the individual documents out. After the documents are separated out into separate electronic files, those documents can be more efficiently and easily docketed. Deadlines and tasks can be prepared and assigned based on the individual documents. The methods and systems discussed herein can allow for more efficient and effective docketing of incoming individual documents.

[0007] For example, a method can include receiving an electronic communication with a document, the document containing multiple items to be docketed; identifying each of the multiple items to be docketed; separating each of the multiple items to be docketed to multiple individual items to be docketed; and sending each of the individual items to a docketing system for docketing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. Like numerals having different letter suffixes may represent

different instances of similar components. The drawings illustrate generally, by way of example, but not by way of limitation, various embodiments discussed in the present document.

[0009] FIG. 1 is a schematic diagram of a docketing system in an example.

[0010] FIG. 2 illustrates sample third party data sources that provide docketing data input for an automated docketing system implemented for managing patent portfolios in a sample embodiment.

[0011] FIG. 3 is a schematic diagram of a system including a splitting tool in an example.

[0012] FIG. 4 is a flow chart depicting a method of splitting incoming docketing items in an example.

[0013] FIG. 5 illustrates a block diagram of an example computing machine upon which any one or more of the techniques or methodologies discussed herein may be implemented.

DETAILED DESCRIPTION

[0014] The present disclosure describes, among other things, a document splitting tool for incoming omnibus files. For example, when a PDF file is received, it may contain multiple different items or documents. The document splitting tool can be used to analyze the PDF, tag different items or documents, and initiate splitting of those items into separate files for easy access and use. In some cases, the splitting tool can additionally sort, label, or otherwise direct the split items and documents.

[0015] As used herein, “electronic communication” refers to an electronic message or a method of exchanging messages between people using electronic devices.

[0016] As used herein, “application” or “program” can include a program or piece of software designed and written to fulfill a particular purpose of the user, such as a database application.

[0017] As used herein, “associate” can include a partner or colleague in business or at work, either internal or external.

[0018] As used herein, “unstructured text” or “unstructured data” refers to data that is not organized in a standard format, for example, text in the body of an electronic communication.

[0019] As used herein, “structured text” or “structured data” refers to data that is organized in a standard format such that a recipient may read the data and institute an automated computing system action without human interpretation of the data.

[0020] As used herein, “scraping,” “web scraping,” “data scraping,” or “web crawling,” can refer to automatically mining or collecting data or information, such as from a database or from the internet.

[0021] As used herein, “file” or “matter” can refer to a particular project, enterprise, or undertaking being worked on by an individual or a collaborative group, planned and designed to achieve a particular aim.

[0022] As used herein, “official record,” or “file history,” can refer to data about a file or matter denoting evidence about past events or tasks within that file or matter, such as an electronic record of previous events in the file or matter. An “official record” can be stored with and maintained by an overseeing agency or organization, such as a governmental organization.

[0023] As used herein, “database,” can refer to a structured set of data, such as held in a computer or on the internet, that can be accessible in various ways.

[0024] As used herein, “splitting,” can refer to separating various files into individual documents or files.

[0025] As used herein, “omnibus document” can refer to a single electronic file containing more than one item. For example, an omnibus document may be an electronic file containing two or more documents.

[0026] FIG. 1 is a block diagram of a docketing system **100** in an example. The docketing system **100** can be automated or semi-automated. The docketing system **100** can include docketing data input **105**, docketing manager **110**, data extraction **115**, auxiliary annotation system **120**, automated docketing using annotations **125**, Universal Procedures Database **130**, reporting tool **135**, customer docketing system **140**, verification system **145**, and machine learning **150**.

[0027] The automated docketing system **100** can receive documents from third party sources including third party docketing systems and/or customer data as docketing data input **105**. The docketing manger system **110** can process the received documents to provide to the customer docketing systems **140** and prepare the documents for data extraction by the data extraction system **115** as needed.

[0028] The data extraction system **115** can perform Optical Character Recognition (OCR) on the received documents from the docketing manger system **110** to extract data, read checkboxes, extract lists, and identify documents where possible. The docketing manger system **110** can also integrate with a Universal Procedures Database (UPDB) **130** to provide automated docketing by an automated docketing tool **125** that processes received documents based on the additional annotations added to the documents based on the complex data extraction performed by the Auxiliary Annotation System (AAS) **120**.

[0029] The AAS **120** may further identify the received documents without using an OCR. To manage this process, the docketing manger system **110** can receive frequent updates of docketing procedure rules including configuration data and updates the UPDB **130** with universal procedure codes (UPCs) as appropriate. The UPCs can be used in conjunction with customer specific codes, checklists, and templates. The rules can specify how to fill in the templates and how to complete customer-specific procedures such as how to docket documents into the customer's docketing system **140**, for example. The template can be filled out by pulling in attributes from the annotations in a document.

[0030] The docketing manger system **110** can receive or intake documents and docketing data from several different sources of docketing data input **105**, validate the docketing items against entries in a customer's docketing system **140**, and communicate those documents to the customer's docketing system **140** via a unified interface. The docketing manger system **110** can also route documents and associated docketing data through the data extraction system **115** and the AAS **120** and organizes the returned metadata and annotations. The docketing manger system **110** thus can provide a breakout between the metadata and the document text.

[0031] The docketing manger system **110** can also keep records and communicate with third-party application programming interfaces (APIs) to push the docketing data and documents automatically where allowed. Otherwise, the docketing manger system **110** can present the documents to human docketers to docket. The docketing manger system **110** may also issue reports upon request.

[0032] The Docketing manger system **110** can be integrated with a customer's existing docketing system (e.g., Foundation IP), semi-integrated (e.g., CPI, Anaqua, etc.), may provide a virtual host that does not talk at all to the customer's existing docketing system (e.g., Lecopio, IP Manager, Memotech), or may provide outputs in spreadsheet form for use by a docketing administrator to update the customer's docking system **140**.

[0033] If the Docketing manger system **110** and the customer's docketing system are not integrated, the data output of automated docketing system **100** may be presented to a human docketer for manual entry. For example, the human docketer may implement macros that interface with the customer's docketing system **140** to populate the received data into the customer's docketing system **140**.

[0034] On the other hand, if the docketing manger system **110** and the customer's docketing system **140** are integrated or semi-integrated, the data output may be processed to determine if any data is missing to automate the docketing process. If anything is missing, the human docketer can add that information before the automated docketing process may proceed further or the data may be auto-populated and mapped to the template from the UPDB **130**.

[0035] The automated docketing system **100** can also perform several post-docketing actions, such as sending docketing reports/details to an external verification system **145** that use a set of rules to verify proper docketing in a host system. The verification system **145** can verify that the data is correctly added to the external customer's docketing system **140**. For example, the verification

system **145** can pull data from the AAS **120**, the docketing manger system **110**, and the customer's docketing system **140** to compare what is present to what is expected to be present in the respective systems.

[0036] The automated docketing system **100** may also provide automated email “report out” notifications to customers by implementing a reporting tool **135** that specifies docketing actions based on UPDB template configurations. The reporting tool **135** can also provide completed docketing reports to customers either directly or via the customers' docketing system **140**.

[0037] In some cases, machine learning techniques may be used to generate annotations. For example, a database of past documents that have been identified may be provided by the docketing manger system **110** and used as a data warehouse to train and improve machine learning models by creating a training set for the machine learning model. Over time, the machine learning model system **150** can learn which PTO IDs to use for which documents, which document in a bundle of documents may be used to characterize the bundle, and may provide predicted PTO IDs for the received documents. The machine learning model system **150** can also establish rule engine prediction capabilities for received documents that test the classifications.

[0038] FIG. 2 illustrates sample third party data sources that provide docketing data input **105** for an automated docketing system **100** implemented for managing patent portfolios in an example. As illustrated in FIG. 2, the third party data sources may include the Patent Office (e.g., USPTO) docketing portal **200**, which provides documents from the USPTO in portable document format (PDF) and includes metadata identifying the title, document code, and mail date for the corresponding document. The third party data sources may further include USPTO PAIR extensible markup language (XML) files **210**, which provide documents from the USPTO in PDF and includes an XML file for patent file wrappers. The third party data sources may also include foreign agents **220** who provide emails with attachments and optional metadata. Foreign agents **220** may also provide hard copy documents that may be scanned for data entry. Similarly, law firms and/or corporate law departments **230** may provide emails with attachments and optional metadata as well as hard copy documents that may be scanned for data entry. Also, third party docketing systems **240** may provide real-time or batch extracts of data for entry into a docketing management system.

[0039] FIG. 3 is a schematic diagram of a file management system **300** with a splitting tool **350** in an example. The system **300** can include an electronic communication system **310**, a file database **320**, an intake tool **330**, a splitting tool **350**, a docketing system **360**, and a file record **370**.

[0040] The electronic communication system **310** can be, for example, an e-mail, text, audio, or other means of communicating with internal and external personnel and entities. The electronic communication system **310** can be a user-accessible system for receiving and/or sending messages, such as through a user interface or other computer. The electronic communication system **310** can be, for example, an e-mail server or other communication system.

[0041] The file database **320** can include a repository of files or projects being working on by the company. The file database **320** can be, for example, a public or private database, such as a governmental run database. In an example, the USPTO PAIR database can be accessed. In an example, ESPACENET can be accessed. In some cases, other database can be used. In some cases, privately run and updated databases can be accessed. The file database **320** can include information on various files of interest for the business. For example, the file database **320** can include information about patent applications or trademark applications. Such information can, for example, include communications from a governmental agency, responses filed by the company, and other official documents.

[0042] The intake tool **330** can include a program or application for receiving electronic communications and associated documents or files. In some cases, the intake tool **330** can be configured to analyze incoming documents to determine if an omnibus file needing splitting is present. In some cases, the intake tool **330** can flag an omnibus file for splitting for presentation to

a user. In some cases, the intake tool **330** can automatically send the omnibus document to the spitting tool **350**. In some cases, the intake tool **330** can be a user interface for interaction with a user. In this case, the user can identify and select omnibus files to send to the splitting tool **350**. [0043] The splitting tool **350** can be an application or program for separating multiple items included in an omnibus file. The splitting tool **350** can receive one or more omnibus files from the intake tool **330** and/or a user.

[0044] The splitting tool **350** can be trained, for example by machine learning, to recognize one or more phrases, characters, or codes, to aid in identifying where or how the omnibus document should be split into multiple documents. The splitting tool **350** can analyze the received omnibus file, such as by an auxiliary annotation system (AAS), optical character recognition (OCR), or other character or image recognition techniques, such as discussed above with reference to the docketing system in FIG. **1**. After analyzing and identifying one or more locations where the omnibus document should be split, the splitting tool can separate the omnibus document into several files at the identified locations.

[0045] The docketing system **360** can be an automated or semi-automated docketing system, such as the docketing system discussed above with reference to FIG. **1**. The docketing system **360** can be in communication with the intake tool **330** and the splitting tool **350**, and can receive documents, tasks, and communications with the intake tool **330**. The docketing system **360** can communicate with and update the file records **370**.

[0046] The file records **370** can, for example, be a local or cloud based file storage system including information of files and projects being worked on at or monitored by the company. The file records **370** can contain historical records, such as past events, communications, and decisions in each file.

[0047] FIG. **4** is a flow chart depicting a method **400** of splitting incoming docketing items in an example. The method **400** can include blocks **410** to **440**.

[0048] At block **410**, an electronic communication is received with an omnibus document. The electronic communication can be, for example, an e-mail, a text, or other type of communication including a document, such as an attachment, an embedded file, or a link to a cloud document.

[0049] At block **420**, an omnibus document is identified, and multiple items within the omnibus documents are identified. An omnibus document can be a document containing two or more discrete items, such as events or files. An intake tool application can be used to identify omnibus documents, such as by OCR. The intake tool can be trained, such as by machine learning, to identify codes, phrase, or characters indicating more than one item in the omnibus document. The intake tool can use a system, such as an AAS, to apply one or more annotations to the omnibus document, indicating the document requires splitting. The omnibus document can then either manually or automatically be sent for splitting.

[0050] An application or program, such as a splitting tool, can then receive the identified omnibus document, and analyze it. For example, the splitting tool can conduct OCR to identify one or more codes, phrases, document types, or events, within the omnibus document. The splitting tool can flag individual files within the document based on this analysis, such as by adding annotations to the omnibus file through an AAS system.

[0051] At block **430**, each of the multiple items is separated. This can be done, for example, by automated or semi-automated action at the spitting tool. For example, the omnibus file can contain one or more annotations (e.g., structured text identifiable by the system) indicating a split should be made at a particular point within the file. The splitting tool can accordingly slice up the document into multiple different documents, each representing one item or event. The splitting tool can confirm successful splitting by additional review, either automated or manual.

[0052] At block **440**, each of the multiple items is sent to be docketed. Once splitting is complete, the splitting tool can send the individual items out to a docketing system such as that shown in FIG.

[0053] FIG. 5 illustrates a block diagram of an example computing system machine **700** upon which any one or more of the techniques (e.g., methodologies) discussed herein may perform. Machine **700** (e.g., computer system) may include a hardware processor **702** (e.g., a central processing unit (CPU), a graphics processing unit (GPU), a hardware processor core, or any combination thereof), a main memory **704** and a static memory **706**, connected via an interconnect **708** (e.g., link or bus), as some or all of these components may constitute hardware for systems **100** or **200** or hardware to operate the services and subsystems and related implementations discussed above.

[0054] Specific examples of main memory **704** include Random Access Memory (RAM), and semiconductor memory devices, which may include, in some embodiments, storage locations in semiconductors such as registers. Specific examples of static memory **706** include non-volatile memory, such as semiconductor memory devices (e.g., Electrically Programmable Read-Only Memory (EPROM), Electrically Erasable Programmable Read-Only Memory (EEPROM)) and flash memory devices; magnetic disks, such as internal hard disks and removable disks; magneto-optical disks; RAM; and CD-ROM and DVD-ROM disks.

[0055] The machine **700** may further include a display device **710**, an input device **712** (e.g., a keyboard), and a user interface (UI) navigation device **714** (e.g., a mouse). In an example, the display device **710**, input device **712** and UI navigation device **714** may be a touch screen display. The machine **700** may additionally include a mass storage device **716** (e.g., drive unit), a signal generation device **718** (e.g., a speaker), a network interface device **720**, and one or more sensors **730**, such as a global positioning system (GPS) sensor, compass, accelerometer, or some other sensor. The machine **700** may include an output controller **728**, such as a serial (e.g., universal serial bus (USB), parallel, or other wired or wireless (e.g., infrared (IR), near field communication (NFC), etc.) connection to communicate or control one or more peripheral devices (e.g., a printer, card reader, etc.). In some embodiments the hardware processor **702** and/or instructions **724** may comprise processing circuitry and/or transceiver circuitry.

[0056] The mass storage device **716** may include a machine readable medium **722** on which is stored one or more sets of data structures or instructions **724** (e.g., software) embodying or utilized by any one or more of the techniques or functions described herein. The instructions **724** may also reside, completely or at least partially, within the main memory **704**, within static memory **706**, or within the hardware processor **702** during execution thereof by the machine **700**. In an example, one or any combination of the hardware processor **702**, the main memory **704**, the static memory **706**, or the mass storage device **716** constitutes, in at least some embodiments, machine readable media.

[0057] The term “machine readable medium” includes, in some embodiments, any medium that is capable of storing, encoding, or carrying instructions for execution by the machine **700** and that cause the machine **700** to perform any one or more of the techniques of the present disclosure, or that is capable of storing, encoding or carrying data structures used by or associated with such instructions. Specific examples of machine readable media include, one or more of non-volatile memory, such as semiconductor memory devices (e.g., Electrically Programmable Read-Only Memory (EPROM), Electrically Erasable Programmable Read-Only Memory (EEPROM)) and flash memory devices; magnetic disks, such as internal hard disks and removable disks; magneto-optical disks; RAM; and CD-ROM and DVD-ROM disks. While the machine readable medium **722** is illustrated as a single medium, the term “machine readable medium” includes, in at least some embodiments, a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) configured to store the one or more instructions **724**. In some examples, machine readable media includes non-transitory machine readable media. In some examples, machine readable media includes machine readable media that is not a transitory propagating signal.

[0058] The instructions **724** are further transmitted or received, in at least some embodiments, over

a communications network **726** using a transmission medium via the network interface device **720** utilizing any one of a number of transfer protocols (e.g., frame relay, internet protocol (IP), transmission control protocol (TCP), user datagram protocol (UDP), hypertext transfer protocol (HTTP), etc.). Example communication networks include a local area network (LAN), a wide area network (WAN), a packet data network (e.g., the Internet), mobile telephone networks (e.g., cellular networks), Plain Old Telephone (POTS) networks, and wireless data networks (e.g., Institute of Electrical and Electronics Engineers (IEEE) 802.11 family of standards known as Wi-Fi®, IEEE 802.15.4 family of standards, a Long Term Evolution (LTE) 4G or 5G family of standards, a Universal Mobile Telecommunications System (UMTS) family of standards, peer-to-peer (P2P) networks, satellite communication networks, among others.

[0059] An apparatus of the machine **700** includes, in at least some embodiments, one or more of a hardware processor **702** (e.g., a central processing unit (CPU), a graphics processing unit (GPU), a hardware processor core, or any combination thereof), a main memory **704** and a static memory **706**, sensors **730**, network interface device **720**, antennas **732**, a display device **710**, an input device **712**, a UI navigation device **714**, a mass storage device **716**, instructions **724**, a signal generation device **718**, and an output controller **728**. The apparatus is configured, in at least some embodiments, to perform one or more of the methods and/or operations disclosed herein. The apparatus is, in some examples, a component of the machine **700** to perform one or more of the methods and/or operations disclosed herein, and/or to perform a portion of one or more of the methods and/or operations disclosed herein.

[0060] In an example embodiment, the network interface device **720** includes one or more physical jacks (e.g., Ethernet, coaxial, or phone jacks) or one or more antennas to connect to the communications network **726**. In an example embodiment, the network interface device **720** includes one or more antennas **732** to wirelessly communicate using at least one of single-input multiple-output (SIMO), multiple-input multiple-output (MIMO), or multiple-input single-output (MISO) techniques. In some examples, the network interface device **720** wirelessly communicates using Multiple User MIMO techniques. The term “transmission medium” shall be taken to include any intangible medium that is capable of storing, encoding or carrying instructions for execution by the machine **700**, and includes digital or analog communications signals or other intangible medium to facilitate communication of such software.

[0061] At least some example embodiments, as described herein, include, or operate on, logic or a number of components, modules, or mechanisms. Such components are tangible entities (e.g., hardware) capable of performing specified operations and are configured or arranged in a certain manner. In an example, circuits are arranged (e.g., internally or with respect to external entities such as other circuits) in a specified manner as a module. In an example, the whole or part of one or more computer systems (e.g., a standalone, client or server computer system) or one or more hardware processors are configured by firmware or software (e.g., instructions, an application portion, or an application) as a component that operates to perform specified operations. In an example, the software resides on a machine readable medium. In an example, the software, when executed by the underlying hardware of the component, causes the hardware to perform the specified operations.

[0062] Accordingly, such components are understood to encompass a tangible entity, be that an entity that is physically constructed, specifically configured (e.g., hardwired), or temporarily (e.g., transitorily) configured (e.g., programmed) to operate in a specified manner or to perform part or all of any operation described herein. Considering examples in which components are temporarily configured, each of the modules need not be instantiated at any one moment in time. For example, where the components comprise a general-purpose hardware processor configured using software, in some embodiments, the general-purpose hardware processor is configured as respective different components at different times. Software accordingly configures a hardware processor, for example, to constitute a particular component at one instance of time and to constitute a different component

at a different instance of time.

[0063] Some embodiments are implemented fully or partially in software and/or firmware. This software and/or firmware takes the form of instructions contained in or on a non-transitory computer-readable storage medium, in at least some embodiments. Those instructions are then read and executed by one or more hardware processors to enable performance of the operations described herein, in at least some embodiments. The instructions are in any suitable form, such as but not limited to source code, compiled code, interpreted code, executable code, static code, dynamic code, and the like. Such a computer-readable medium includes any tangible non-transitory medium for storing information in a form readable by one or more computers, such as but not limited to read only memory (ROM); random access memory (RAM); magnetic disk storage media; optical storage media; flash memory, etc.

[0064] Various embodiments may be implemented fully or partially in software and/or firmware. This software and/or firmware may take the form of instructions contained in or on a non-transitory computer-readable storage medium. Those instructions are then read and executed by one or more processors to enable performance of the operations described herein. The instructions are in any suitable form, such as but not limited to source code, compiled code, interpreted code, executable code, static code, dynamic code, and the like. Such a computer-readable medium includes, in at least some embodiments, any tangible non-transitory medium for storing information in a form readable by one or more computers, such as but not limited to read only memory (ROM); random access memory (RAM); magnetic disk storage media; optical storage media; flash memory, etc.

VARIOUS NOTES & EXAMPLES

[0065] Example 1 can include a method comprising: receiving an electronic communication with a document, the document containing multiple items to be docketed; identifying each of the multiple items to be docketed; splitting the document into individual items to be docketed; and sending each of the individual items to a docketing system for docketing.

[0066] Example 2 can include Example 1, further comprising determining one or more docket tasks associated with each of the multiple items to be docketed.

[0067] Example 3 can include any of Examples 1-2, further comprising determining one or more deadlines associated with each of the multiple items to be docketed.

[0068] Example 4 can include any of Examples 1-3, further comprising labeling each of the individual items.

[0069] Example 5 can include any of Examples 1-4, further comprising sorting each of the individual items into one or more categories.

[0070] Example 6 can include any of Examples 1-5, further comprising applying structured text to each of the individual items, the structured text denoting the one or more categories.

[0071] Example 7 can include any of Examples 1-6, further comprising directing each of the individual items to one or more locations.

[0072] Example 8 can include any of Examples 1-7, further comprising flagging at least of the individual items for further review.

[0073] Example 9 can include any of Examples 1-8, further comprising identifying the document containing multiple items to be docketed prior to identifying each of the multiple items to be docketed.

[0074] Example 10 can include any of Examples 1-9, wherein identifying the document containing multiple items comprises optical character recognition.

[0075] Example 11 can include any of Examples 1-10, wherein identifying the document containing multiple items comprises using an auxiliary annotation system and applying one or more annotations to the document.

[0076] Example 12 can include any of Examples 1-11, wherein identifying each of the multiple items to be docketed comprise analyzing the document containing multiple items with optical character recognition.

[0077] Example 13 can include any of Examples 1-12, wherein optical character recognition is used to identify one or more codes, phrases, templates, or other objects indicating different items in the omnibus file.

[0078] Example 14 can include any of Examples 1-13, wherein identifying each of the multiple items to be docketed comprise analyzing the document containing multiple items using an auxiliary annotation system.

[0079] Example 15 can include any of Examples 1-14, further comprising applying one or more annotations to the document indicating individual items therein.

[0080] Each of these non-limiting examples can stand on its own, or can be combined in various permutations or combinations with one or more of the other examples.

[0081] The above detailed description includes references to the accompanying drawings, which form a part of the detailed description. The drawings show, by way of illustration, specific embodiments in which the invention can be practiced. These embodiments are also referred to herein as “examples.” Such examples can include elements in addition to those shown or described. However, the present inventors also contemplate examples in which only those elements shown or described are provided. Moreover, the present inventors also contemplate examples using any combination or permutation of those elements shown or described (or one or more aspects thereof), either with respect to a particular example (or one or more aspects thereof), or with respect to other examples (or one or more aspects thereof) shown or described herein.

[0082] In the event of inconsistent usages between this document and any documents so incorporated by reference, the usage in this document controls.

[0083] In this document, the terms “a” or “an” are used, as is common in patent documents, to include one or more than one, independent of any other instances or usages of “at least one” or “one or more.” In this document, the term “or” is used to refer to a nonexclusive or, such that “A or B” includes “A but not B,” “B but not A,” and “A and B,” unless otherwise indicated. In this document, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Also, in the following claims, the terms “including” and “comprising” are open-ended, that is, a system, device, article, composition, formulation, or process that includes elements in addition to those listed after such a term in a claim are still deemed to fall within the scope of that claim. Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects.

[0084] Method examples described herein can be machine or computer-implemented at least in part. Some examples can include a computer-readable medium or machine-readable medium encoded with instructions operable to configure an electronic device to perform methods as described in the above examples. An implementation of such methods can include code, such as microcode, assembly language code, a higher-level language code, or the like. Such code can include computer readable instructions for performing various methods. The code may form portions of computer program products. Further, in an example, the code can be tangibly stored on one or more volatile, non-transitory, or non-volatile tangible computer-readable media, such as during execution or at other times. Examples of these tangible computer-readable media can include, but are not limited to, hard disks, removable magnetic disks, removable optical disks (e.g., compact disks and digital video disks), magnetic cassettes, memory cards or sticks, random access memories (RAMs), read only memories (ROMs), and the like.

[0085] The above description is intended to be illustrative, and not restrictive. For example, the above-described examples (or one or more aspects thereof) may be used in combination with each other. Other embodiments can be used, such as by one of ordinary skill in the art upon reviewing the above description. The Abstract is provided to comply with 37 C.F.R. § 1.72(b), to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. Also,

in the above Detailed Description, various features may be grouped together to streamline the disclosure. This should not be interpreted as intending that an unclaimed disclosed feature is essential to any claim. Rather, inventive subject matter may lie in less than all features of a particular disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description as examples or embodiments, with each claim standing on its own as a separate embodiment, and it is contemplated that such embodiments can be combined with each other in various combinations or permutations. The scope of the invention should be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

Claims

1. A method comprising: receiving a plurality of electronic communications containing documents associated with a file; analyzing each document associated with the file, and identifying omnibus documents within the plurality of electronic communications, wherein each identified omnibus document contains more than one item, each item requiring docketing individually; applying one or more annotations at identified locations within the omnibus document indicating an end of one item and a start of another item; identifying the one or more annotations; displaying a user interface for interaction with a user, the user interface containing the identified omnibus documents and identified one or more annotations within the identified omnibus documents; receiving a selection of an omnibus document from the user interface for splitting; splitting the selected omnibus document into individual items based on the identified locations within the selected omnibus document; and sending each of the more than one items to a docketing system for docketing.
2. The method of claim 1, further comprising determining one or more docket tasks associated with each item requiring docketing individually.
3. The method of claim 1, further comprising determining one or more deadlines associated with each item requiring docketing individually.
4. The method of claim 1, further comprising labeling each item.
5. The method of claim 1, further comprising sorting each item into one or more categories.
6. The method of claim 5, further comprising applying structured text to each item, the structured text denoting the one or more categories.
7. The method of claim 1, further comprising directing each item to one or more locations.
8. The method of claim 1, further comprising flagging at least one of the more than one items for further review.
9. The method of claim 1, further comprising identifying each of the more than one items to be docketed.
10. The method of claim 9, wherein identifying each of the more than one items to be docketed comprises analyzing the identified omnibus documents using an auxiliary annotation system.
11. The method of claim 9, wherein identifying each of the more than one items to be docketed comprises analyzing the identified omnibus documents with optical character recognition.
12. The method of claim 1, wherein identifying omnibus documents comprises performing optical character recognition.
13. The method of claim 12, wherein performing optical character recognition includes identifying one or more codes, phrases, templates, or other objects indicating different items within each identified omnibus document.
14. The method of claim 1, wherein identifying omnibus documents comprises using an auxiliary annotation system.
15. The method of claim 14, further comprising applying one or more annotations to each of the identified omnibus documents indicating different items therein.
16. A system comprising: an intake tool configured to: receive a plurality of electronic

communications containing documents associated with a file; analyze each document associated with the file, and identify omnibus documents within the plurality of electronic communications, wherein each identified omnibus document contains more than one item, each item requiring docketing individually; and apply one or more annotations at identified locations within the omnibus document indicating an end of one item and a start of another item, a user interface for interaction with a user, the user interface containing the identified omnibus documents and identified one or more annotations within each identified omnibus document, and configured to receive a selection of an identified omnibus document for splitting; a splitting tool configured to identify the one or more annotations and split the selected omnibus document into individual items based on the identified locations within the selected omnibus document; a docketing system, wherein the docketing system is at least partially automated, the docketing system configured to receive each of the more than one items from the splitting tool.

17. The system of claim 16, further comprising an electronic communication system for receiving the plurality of electronic communications and directing the identified omnibus documents to the intake tool.

18. The system of claim 16, further comprising a file database, wherein the intake tool is configured to scrape the plurality of electronic communications from the file database.

19. The system of claim 16, further comprising file records, wherein the docketing system is configured to communicate with and update the file records.

20. The system of claim 16, wherein the intake tool includes an auxiliary annotation system configured to apply one or more annotations to each of the identified omnibus documents, the annotations indicating different items contained therein.
